Introduction to (Digital) Image Processing

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Digital Image Processing

- **Digital Images:**
  - An image may be defined as *f(x,y)*
    - *(x,y):* spatial coordinates
    - *f(x,y):* intensity or gray level at that point
  - x,y, and the intensity are all finite and discrete
- **Digital image processing**
  - To process digital images by means of digital computers
    - Signal processing and computer vision

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Images

- **We can categorize images according to their source, such as electromagnetic energy spectrum, acoustic, ultrasonic, electronic**

- **Principal energy source for images: EM**
  - EM spectrum arranged according to energy of one photon

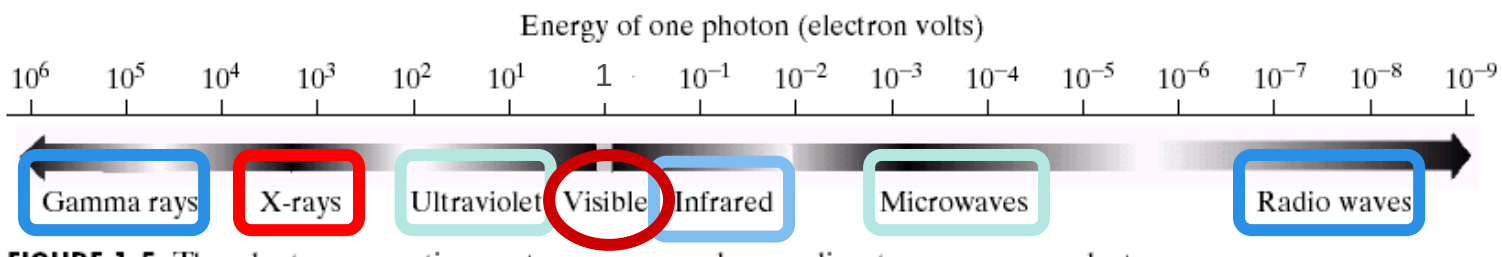

FIGURE 1.5 The electromagnetic spectrum arranged according to energy per photon.

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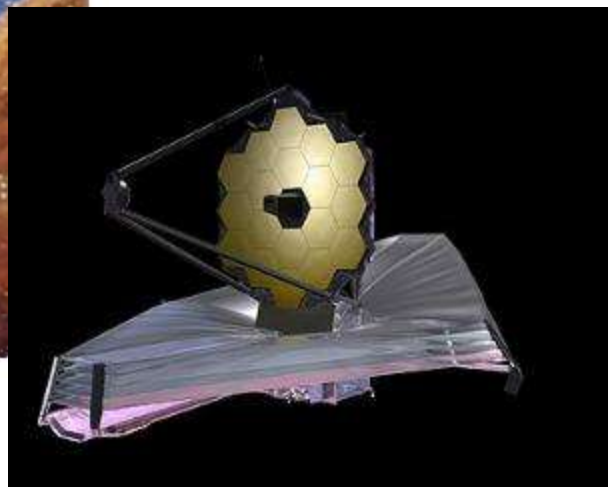
Hubble / James Webb Space Telescopes



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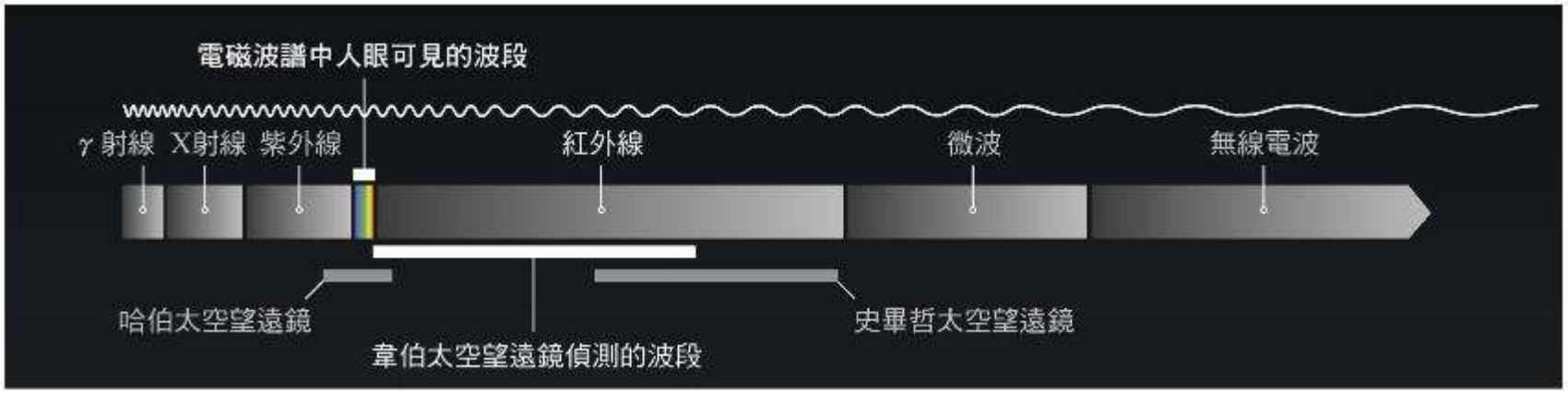
Space Telescopes



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Gamma-Ray Imaging

- **Nuclear medicine and astronomical observations**
  - Radioactive isotope that emits gamma rays
  - PET: positron emission tomography, positron and electron will be annihilated and two gamma rays are given off
  - Natural radiation of the objects

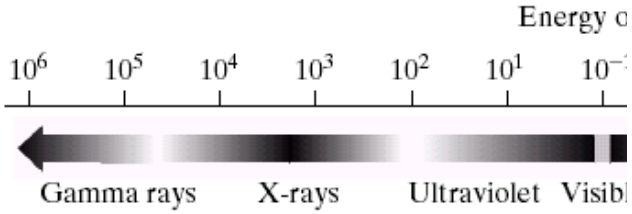

FIGURE 1.5 The electromagnetic spectrum ar

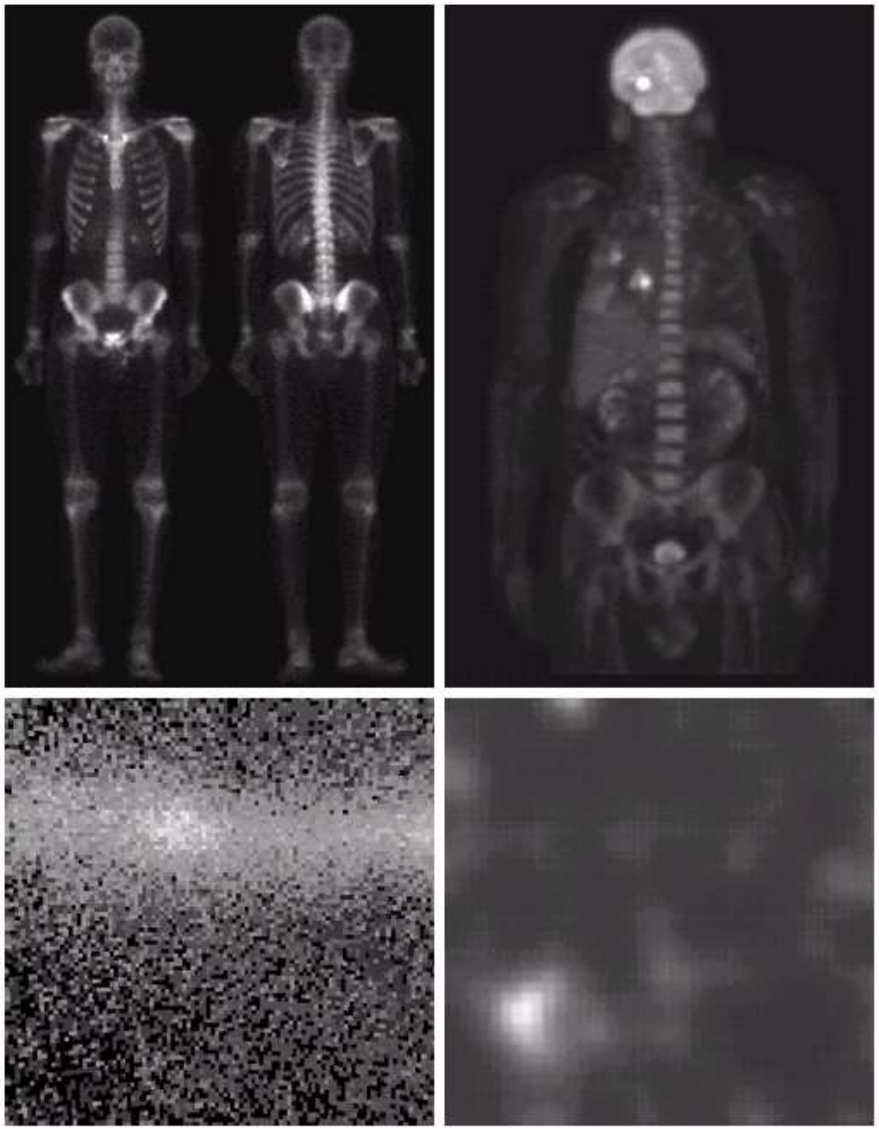

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FIGURE 1.6 Examples of gamma-ray imaging. (a) Bone scan. (b) PET image. (c) Cygnus Loop. (d) Gamma radiation (bright spot) from a reactor valve. (Images courtesy of (a) G.E. Medical Systems, (b) Dr. Michael E. Casey, CTI PET Systems, (c) NASA, (d) Professors Zhong He and David K. Wehe, University of Michigan.)

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X-ray Imaging

- With X-ray tube such that the energy/number is controlled by the voltage and currents
- Computerized axial tomography (CAT)

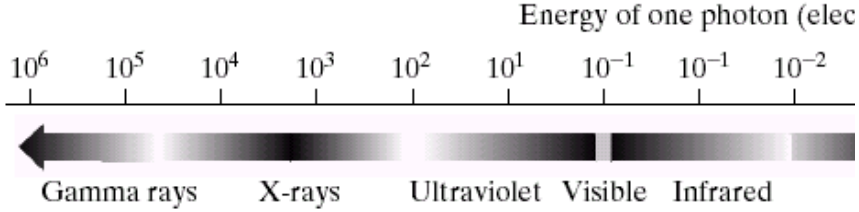

FIGURE 1.5 The electromagnetic spectrum arranged according

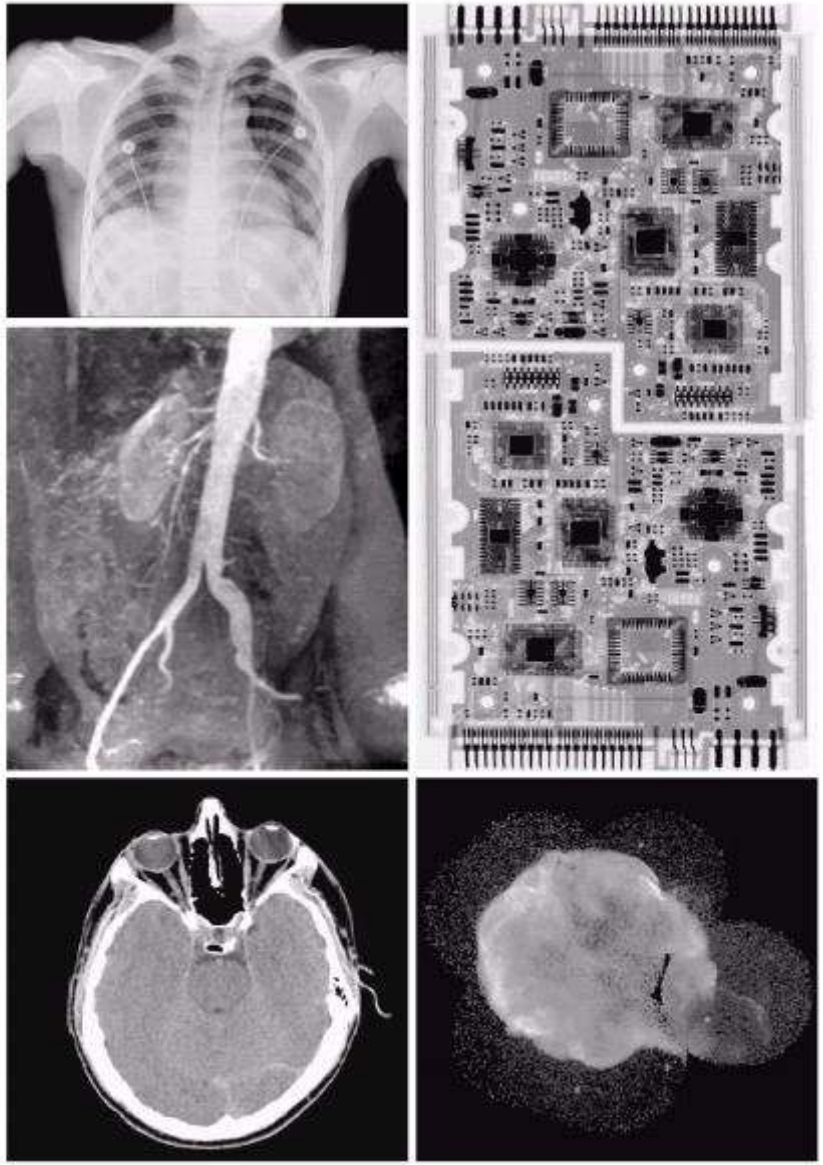

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FIGURE 1.7 Examples of X-ray imaging. (a) Chest X-ray. (b) Aortic angiogram. (c) Head CT. (d) Circuit boards. (e) Cygnus Loop. (Images courtesy of (a) and (c) Dr. David R. Pickens, Dept. of Radiology & Radiological Sciences, Vanderbilt University Medical Center, (b) Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School, (d) Mr. Joseph E. Pascente, Lixi, Inc., and (e) NASA.)

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Ultraviolet Band Imaging

- Ultraviolet light is usually used in fluorescence microscopy
- When a photon of ultraviolet radiation collides with an electron in an atom of a fluorescent material, it elevates to the higher energy and emits light in the form of a lower-energy photon in the visible region

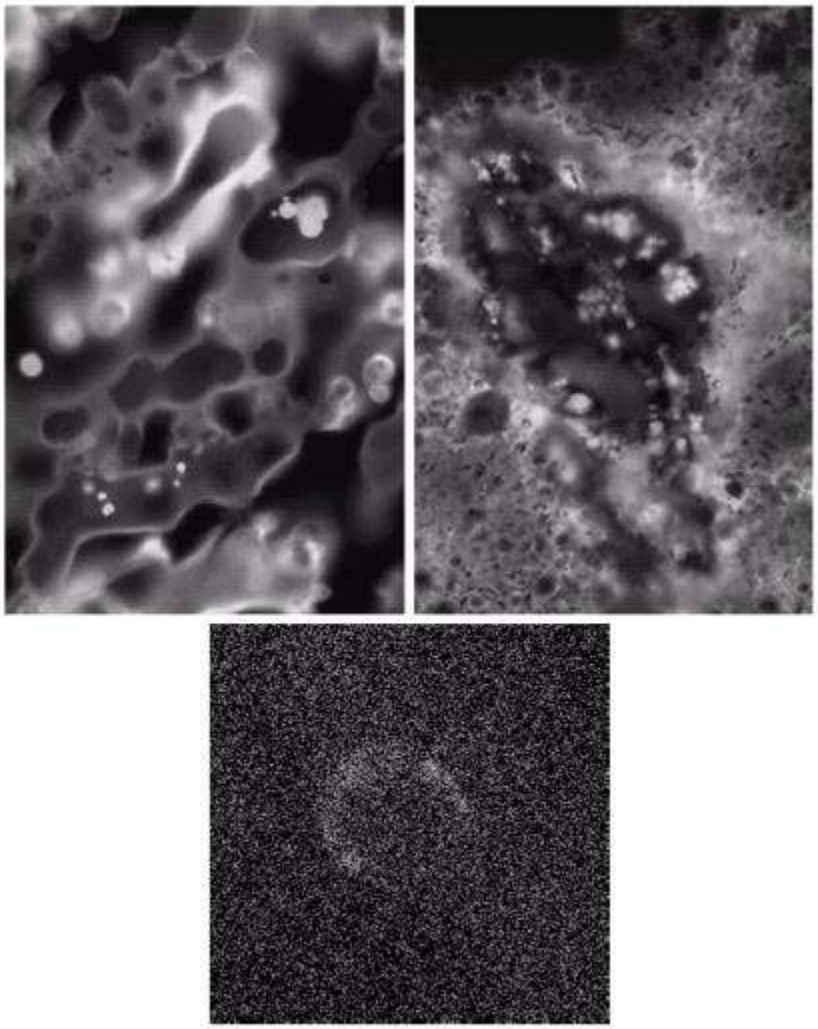

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FIGURE 1.8 Examples of ultraviolet imaging. (a) Normal corn. (b) Smut corn. (c) Cygnus Loop. (Images courtesy of (a) and (b) Dr. Michael W. Davidson, Florida State University, (c) NASA.)

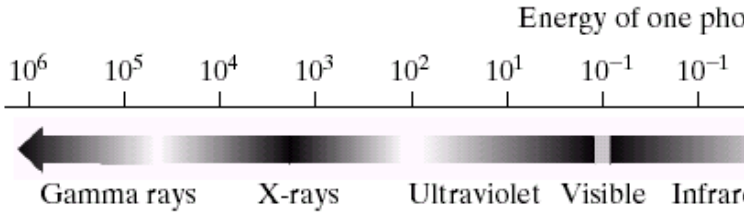

FIGURE 1.5 The electromagnetic spectrum arranged according to energy per photon.

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Visible and Infrared Bands

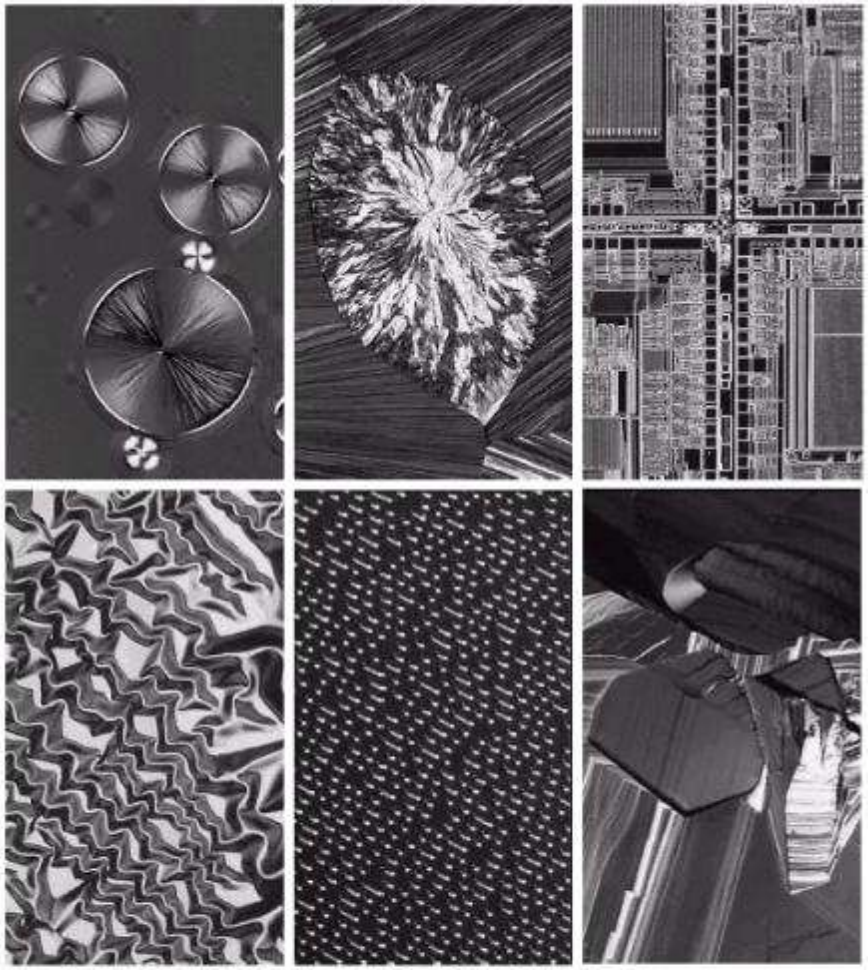

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FIGURE 1.9 Examples of light microscopy images. (a) Taxol (anticancer agent), magnified 250×. (b) Cholesterol—40×. (c) Microprocessor—60×. (d) Nickel oxide thin film—600×. (e) Surface of audio CD—1750×. (f) Organic superconductor—450×. (Images courtesy of Dr. Michael W. Davidson, Florida State University.)

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Visible and Infrared Bands

TABLE 1.1 Thematic bands of NASA's LANDSAT satellite.

Band No.	Name	Wavelength (μm)	Characteristics and Uses
1	Visible blue	0.45–0.52	Maximum water penetration
2	Visible green	0.53–0.61	Measures plant vigor
3	Visible red	0.63–0.69	Vegetation discrimination
4	Near infrared	0.78–0.90	Biomass and shoreline mapping
5	Middle infrared	1.55–1.75	Moisture content: soil/vegetation
6	Thermal infrared	10.4–12.5	Soil moisture; thermal mapping
7	Short-wave infrared	2.09–2.35	Mineral mapping

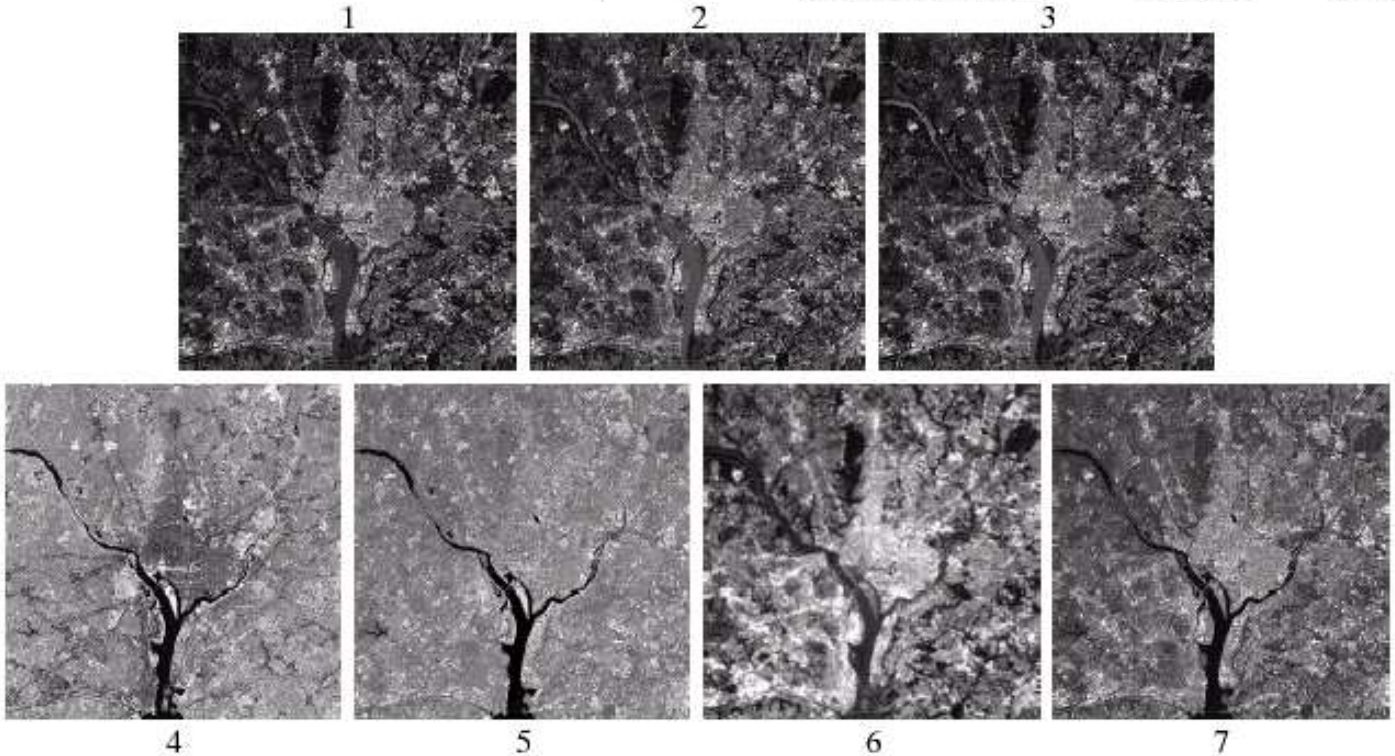

FIGURE 1.10 LANDSAT satellite images of the Washington, D.C. area. The numbers refer to the thematic bands in Table 1.1. (Images courtesy of NASA.)

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Visible and Infrared Bands

FIGURE 1.11 Satellite image of Hurricane Katrina taken on August 29, 2005. (Courtesy of NOAA.)

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Infrared Bands

- 10-13.4 μm

FIGURE 1.12 Infrared satellite images of the Americas. The small shaded map is provided for reference. (Courtesy of NOAA.)

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a b c
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FIGURE 1.14 Some examples of manufactured goods checked using digital image processing. (a) Circuit board controller. (b) Packaged pills. (c) Bottles. (d) Air bubbles in a clear plastic product. (e) Cereal. (f) Image of intraocular implant. (Figure (f) courtesy of Mr. Pete Sites, Perceptics Corporation.)

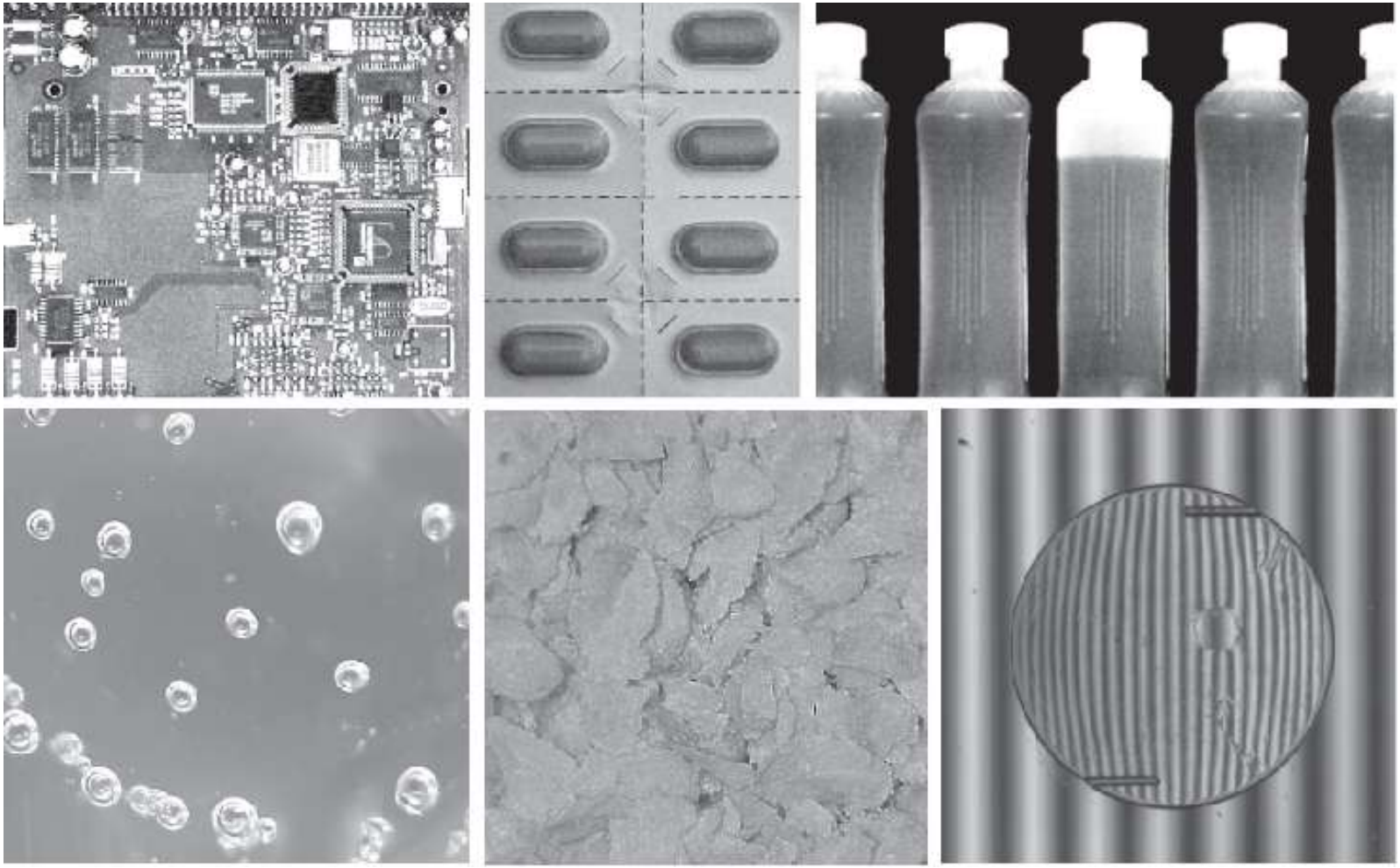

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FIGURE 1.15 Some additional examples of imaging in the visible spectrum. (a) Thumb print. (b) Paper currency. (c) and (d) Automated license plate reading. (Figure (a) courtesy of the National Institute of Standards and Technology. Figures (c) and (d) courtesy of Dr. Juan Herrera, Perceptics Corporation.)

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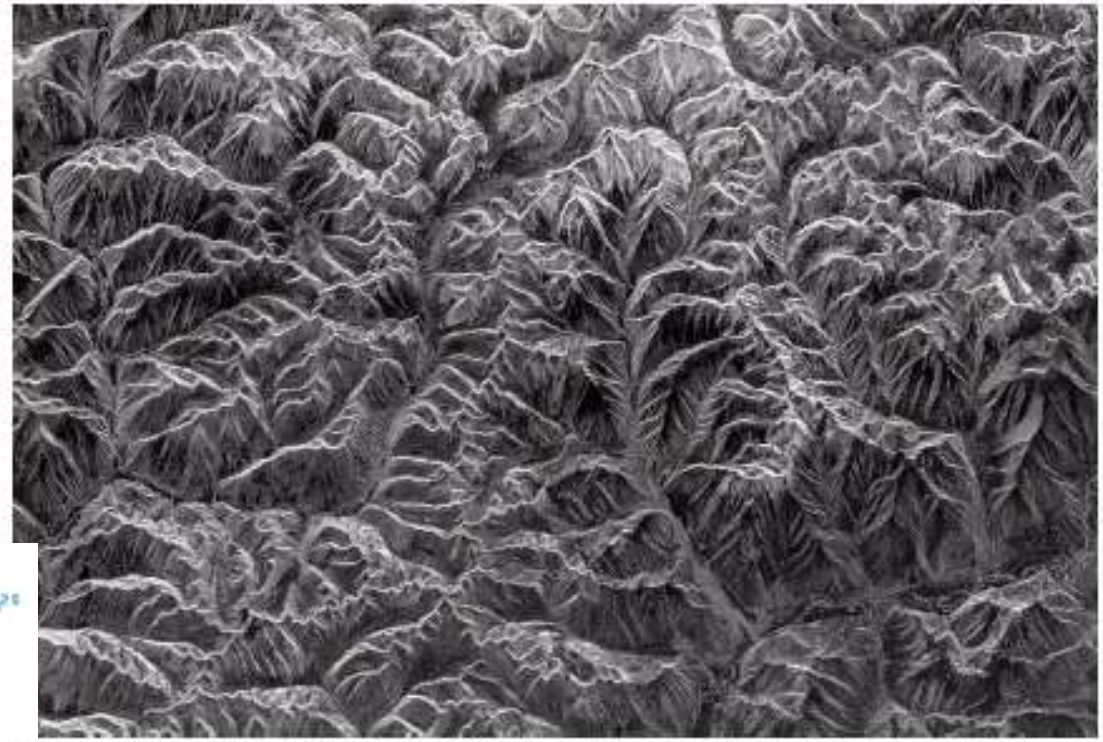
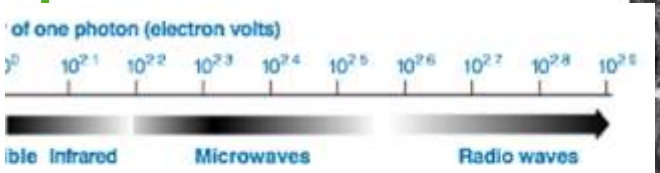
Imaging in the Microwave Band

- The unique feature of imaging radar is its ability to collect data over virtually any region at any time, any weathers, and lighting conditions
- An imaging radar works like a flash camera with its own illumination

FIGURE 1.16 Spaceborne radar image of mountains in southeast Tibet. (Courtesy of NASA.)





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Imaging in the Radio Band MRI: magnetic resonance imaging

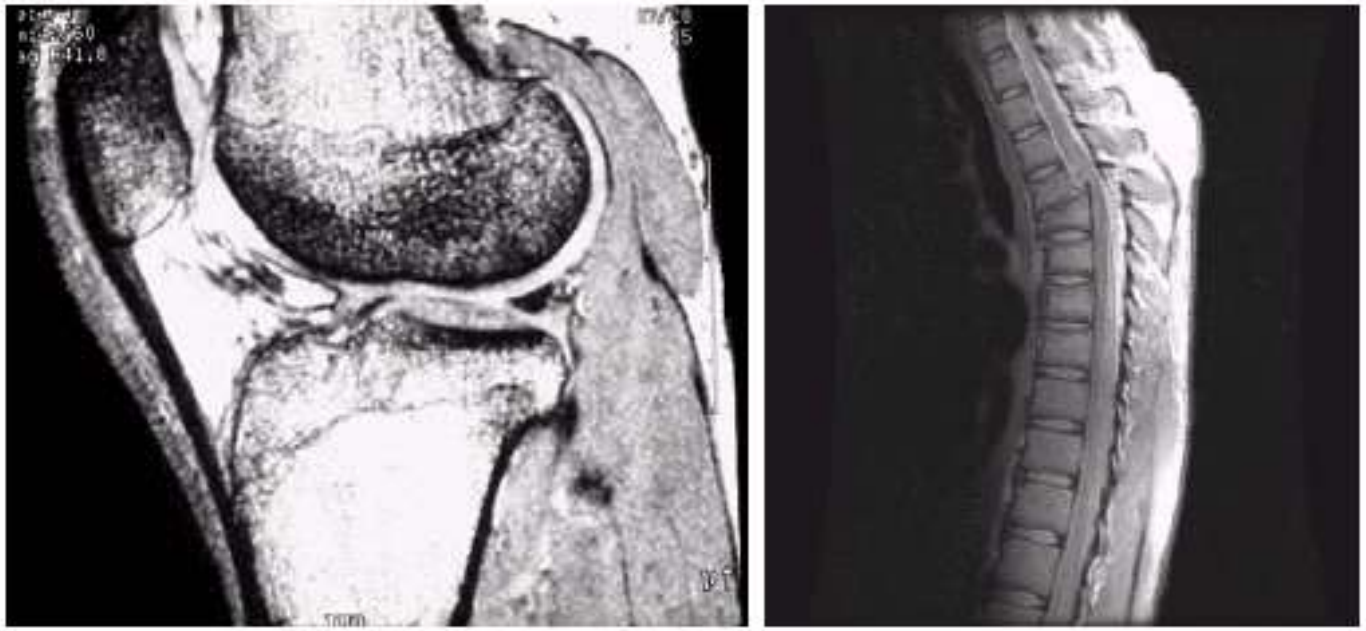

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FIGURE 1.17 MRI images of a human (a) knee, and (b) spine. (Image (a) courtesy of Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School, and (b) Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)

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Imaging in the Radio Band

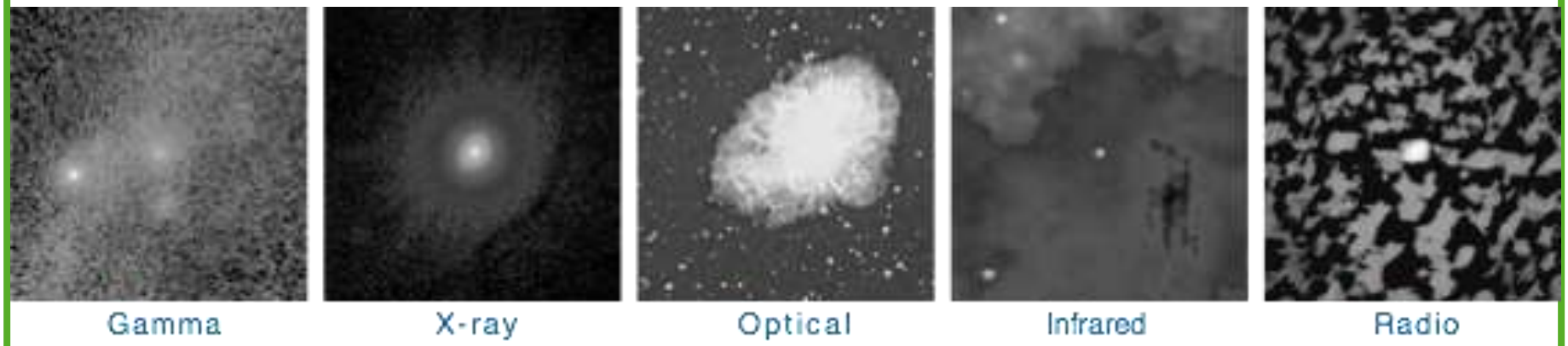

FIGURE 1.18 Images of the Crab Pulsar (in the center of each image) covering the electromagnetic spectrum. (Courtesy of NASA.)

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Ultrasound

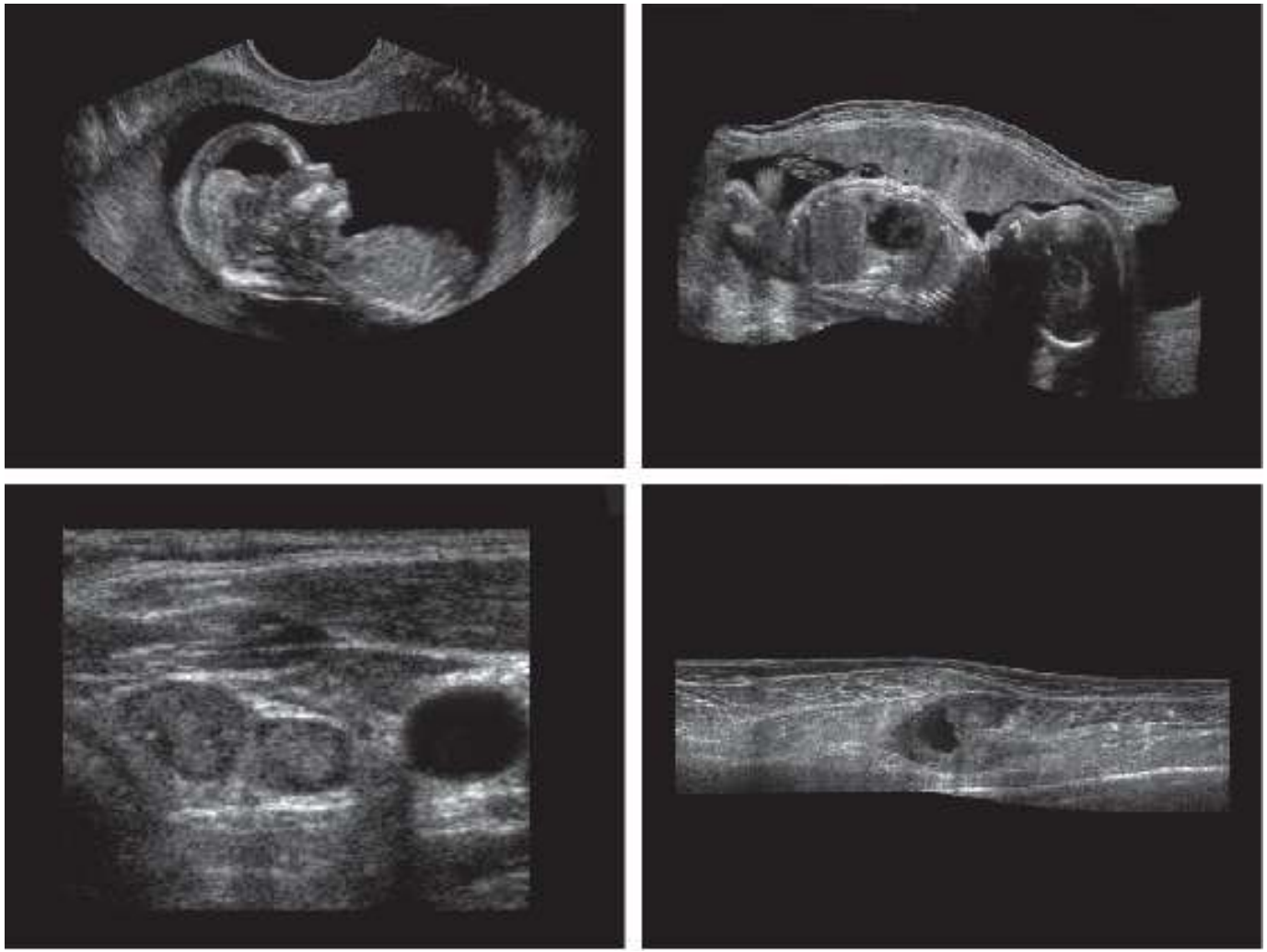

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FIGURE 1.20 Examples of ultrasound imaging. (a) A fetus. (b) Another view of the fetus. (c) Thyroids. (d) Muscle layers showing lesion. (Courtesy of Siemens Medical Systems, Inc., Ultrasound Group.)

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Operations in DIP

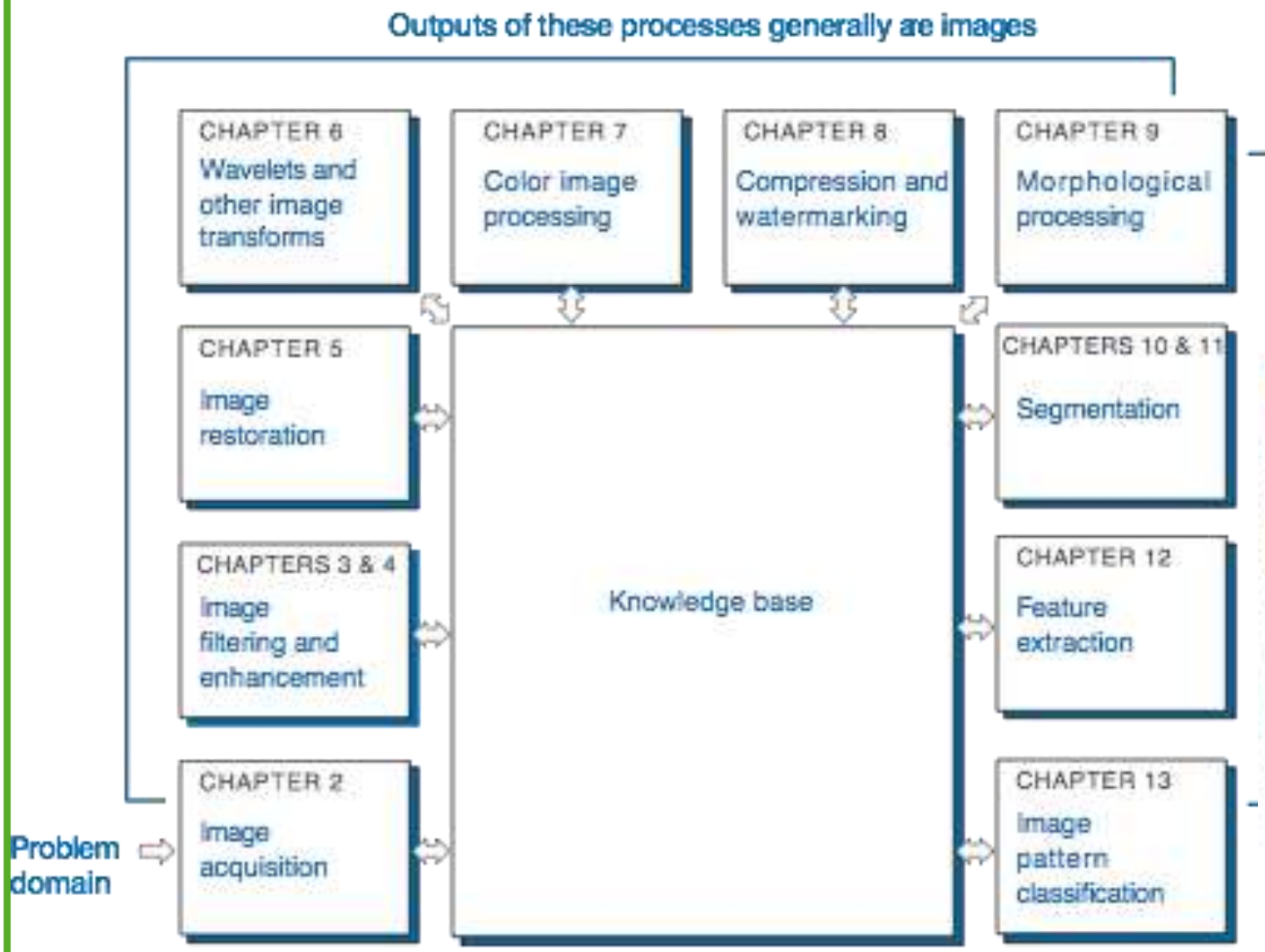

FIGURE 1.23 Fundamental steps in digital image processing. The chapter(s) indicated in the boxes is where the material described in the box is discussed.

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Components of a DIP System

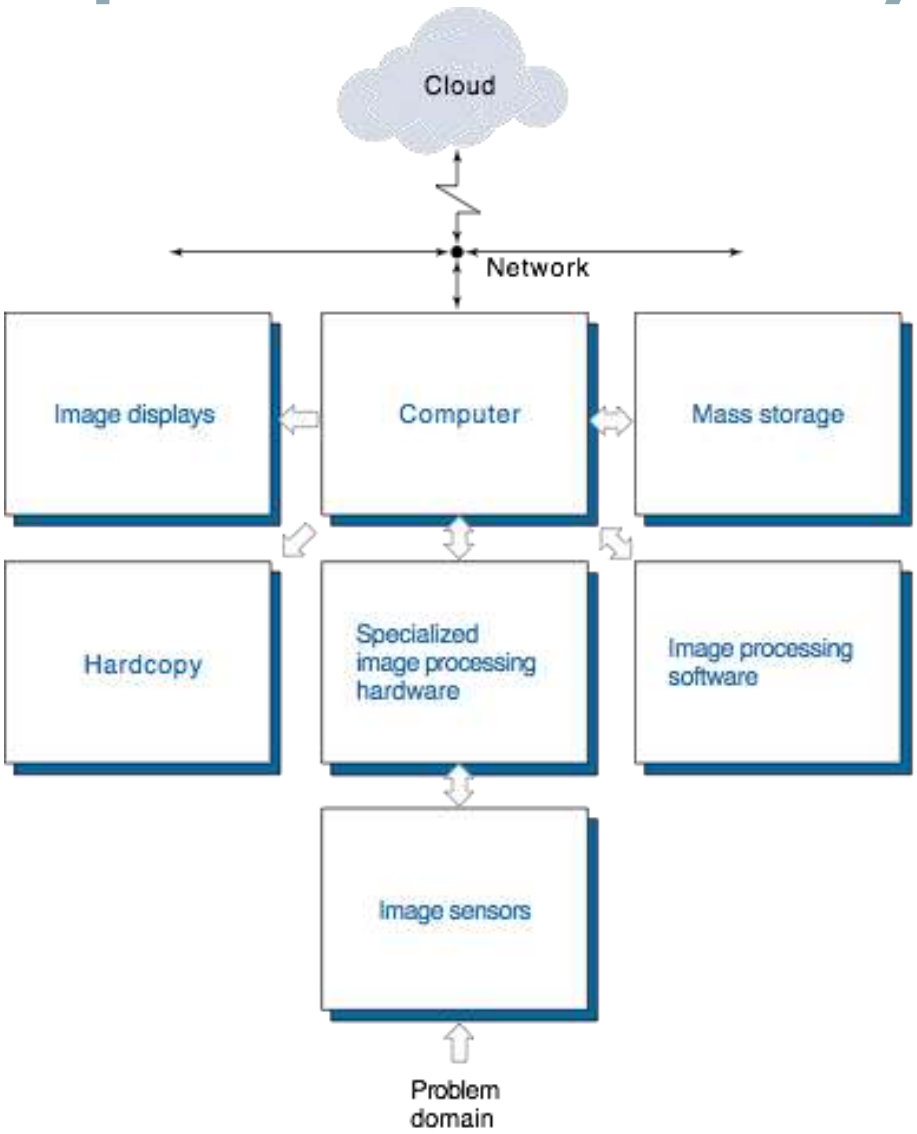

FIGURE 1.24 Components of a general-purpose image processing system.

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